

Hi everyone, I'm Rongjun Huang from ICRAR at UWA.

My work asks a simple question: **does metallicity follow star formation, or fight it?**

Using Virgo spiral galaxies from the MAUVE-MUSE survey, I find that the answer is: **both**.

In the **outer, low- $\Sigma_{\star}$** parts of galaxy discs, regions with higher star formation tend to have **lower gas-phase metallicity**. That suggests fresh gas inflow is feeding star formation and, at the same time, **diluting** the metals.

But in the **inner, high- $\Sigma_{\star}$** regions, the trend flips. There, higher star formation goes together with **higher metallicity**, which points more toward **local enrichment**.

So the key result is that **the same galaxy disc can switch regimes**: from **gas-supply-dominated** in the outer parts to **enrichment-dominated** in the inner parts.

On the plot, the coloured lines are individual galaxies, and the black line is the ensemble trend. You can see the sign flip happens around $\log \Sigma_{\star} \sim 7.5$ to 8.

So, metallicity does not behave the same way everywhere within a galaxy — and that gives us a more resolved view of how star formation, gas cycling, and chemical enrichment are connected.

If you're interested in resolved galaxy evolution in cluster environments, come chat with me. Thank you.